Unit 6.01: Linear Regression & OLS

Q1. A fitted simple linear regression model is hat{y} = 2x + 3 . What is the predicted target value hat{y} when the input feature is x = 4 ?

[x] A: 11 [ ] B: 14
[ ] C: 7 [ ] D: 8

Solution: Substitute x = 4 into the regression equation: hat{y} = 24 + 3 = 8 + 3 = 11 .

Q2. A fitted simple linear regression model is hat{y} = 3x + 6 . What is the predicted target value hat{y} when the input feature is x = 4 ?

[x] A: 18 [ ] B: 21
[ ] C: 14 [ ] D: 12

Solution: Substitute x = 4 into the regression equation: hat{y} = 34 + 6 = 12 + 6 = 18 .

Q3. A fitted simple linear regression model is hat{y} = 4x + 9 . What is the predicted target value hat{y} when the input feature is x = 4 ?

[x] A: 25 [ ] B: 28
[ ] C: 21 [ ] D: 16

Solution: Substitute x = 4 into the regression equation: hat{y} = 44 + 9 = 16 + 9 = 25 .

Q4. A fitted simple linear regression model is hat{y} = 5x + 12 . What is the predicted target value hat{y} when the input feature is x = 4 ?

[x] A: 32 [ ] B: 35
[ ] C: 28 [ ] D: 20

Solution: Substitute x = 4 into the regression equation: hat{y} = 54 + 12 = 20 + 12 = 32 .

Q5. A fitted simple linear regression model is hat{y} = 6x + 15 . What is the predicted target value hat{y} when the input feature is x = 4 ?

[x] A: 39 [ ] B: 42
[ ] C: 35 [ ] D: 24

Solution: Substitute x = 4 into the regression equation: hat{y} = 64 + 15 = 24 + 15 = 39 .

Q6. A fitted simple linear regression model is hat{y} = 7x + 18 . What is the predicted target value hat{y} when the input feature is x = 4 ?

[x] A: 46 [ ] B: 49
[ ] C: 42 [ ] D: 28

Solution: Substitute x = 4 into the regression equation: hat{y} = 74 + 18 = 28 + 18 = 46 .

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Q7. In a multiple linear regression model, the Sum of Squared Errors is text{SSE} = 20 and the Total Sum of Squares is text{SST} = 100 . What is the coefficient of determination R^2 ?

[x] A: R^2 = 0.80
[ ] B: R^2 = 0.20
[ ] C: R^2 = 0.90
[ ] D: R^2 = 0.25

Solution: The coefficient of determination measures the proportion of variance explained: R^2 = 1 - frac{text{SSE}}{text{SST}} = 1 - frac{20}{100} = 1 - 0.20 = 0.80 .

Q8. In a multiple linear regression model, the Sum of Squared Errors is text{SSE} = 40 and the Total Sum of Squares is text{SST} = 200 . What is the coefficient of determination R^2 ?

[x] A: R^2 = 0.80
[ ] B: R^2 = 0.20
[ ] C: R^2 = 0.90
[ ] D: R^2 = 0.25

Solution: The coefficient of determination measures the proportion of variance explained: R^2 = 1 - frac{text{SSE}}{text{SST}} = 1 - frac{40}{200} = 1 - 0.20 = 0.80 .